OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

11.5 Lane linkage

To enable lane navigation, linkage information for lanes may be stored in ASAM OpenDRIVE. Linkage is described by means of `<predecessor>` and `<successor>` elements for each lane. Lanes may be linked to lanes on the same or another road. A `<predecessor>` of a given lane is a lane connected to the start of its lane section in its road reference line direction. A `<successor>` of a given lane is a lane connected to the end of its lane section in road reference line direction. A lane linkage by a `<predecessor>` and a `<successor>` is independent of the driving direction.

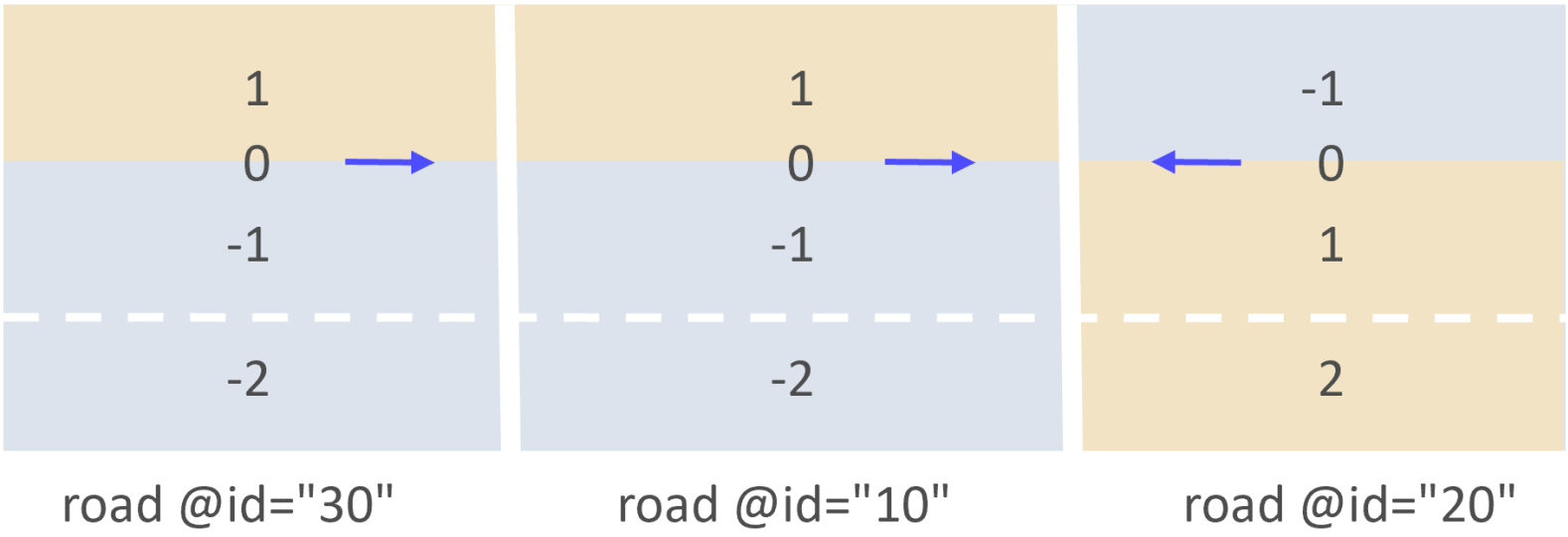

Figure 64. Lane links for road with id 10

Figure 64 shows an example for lane linkage. The considered roads and their predecessor and successor are described in Table 39.

Table 39. Lane predecessors and successors for road with id 10

Considered road	Lane predecessor	Lane successor

Considered road	Lane predecessor	Lane successor
Road 10 with lane id 1	Lane id 1 (Road 30)	Lane id -1 (Road 20)
Road 10 with lane id -1	Lane id -1 (Road 30)	Lane id 1 (Road 20)
Road 10 with lane id -2	Lane id -2 (Road 30)	Lane id 2 (Road 20)

Lane predecessors and successors shall only be used to connect lanes if a physical connection at the beginning or end of both lanes exist. Both lanes have a non-zero width at the connection point and are semantically connected.

Examples where lane linkage should be used:

- asam.net:xodr:1.4.0:road.lane.link.lanes_across_laneSections: Lane that continues across the lane sections shall be connected in both directions.
- Lane changes its type on a highway from `exit` to `offRamp`.
- Parking lane at the roadside ends, the road starts, and a vehicle can continue driving from the parking lane on the driving lane.

Example where lane linkage should not be used:

- Parking lane at the roadside ends and a grass strip begins.

Examples where multiple predecessors and successors shall be used:

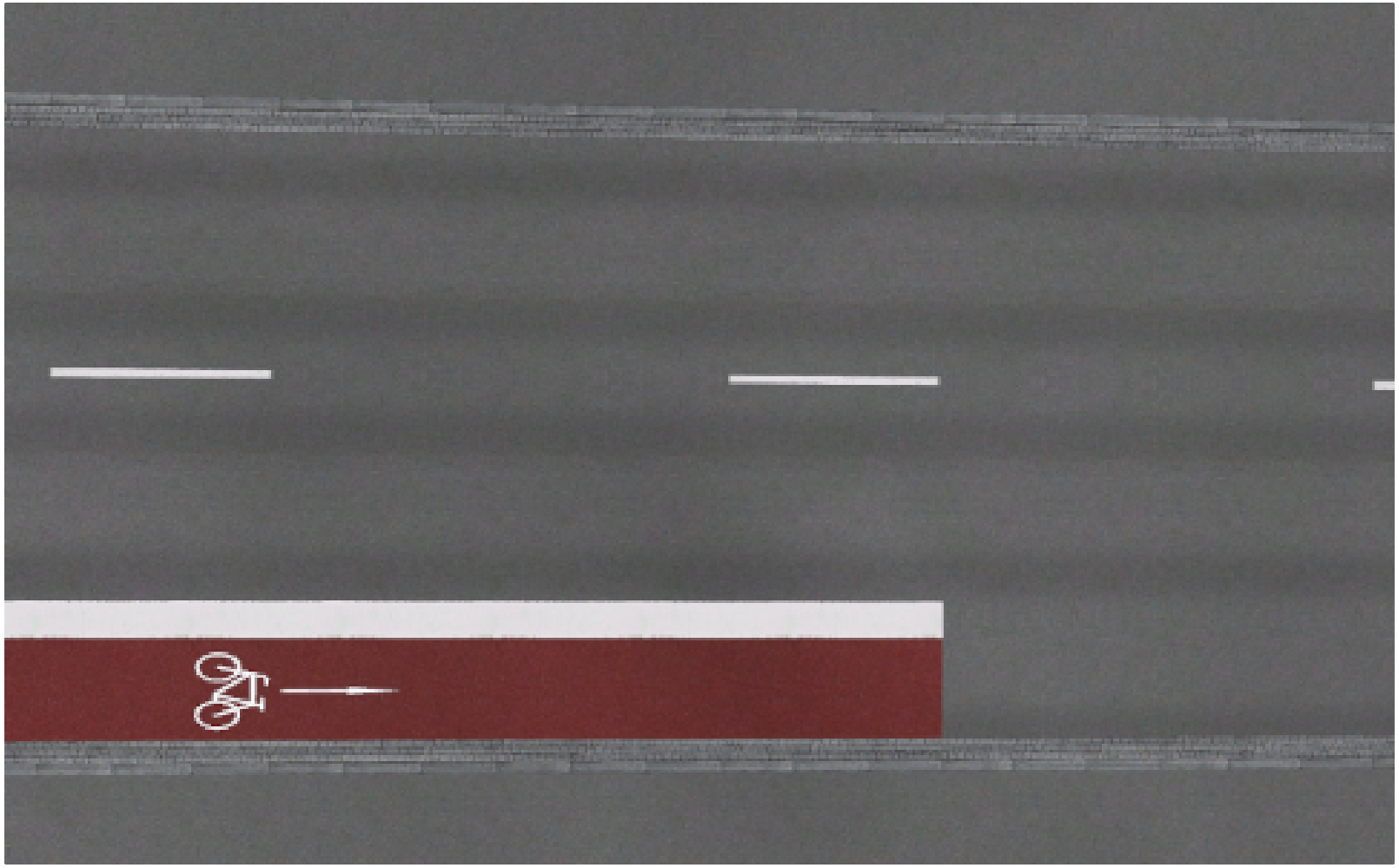

Figure 65. Example of an ending bikeway

Figure 65 shows an example where a bikeway ends and bicycles are expected to continue driving on the driving lane.

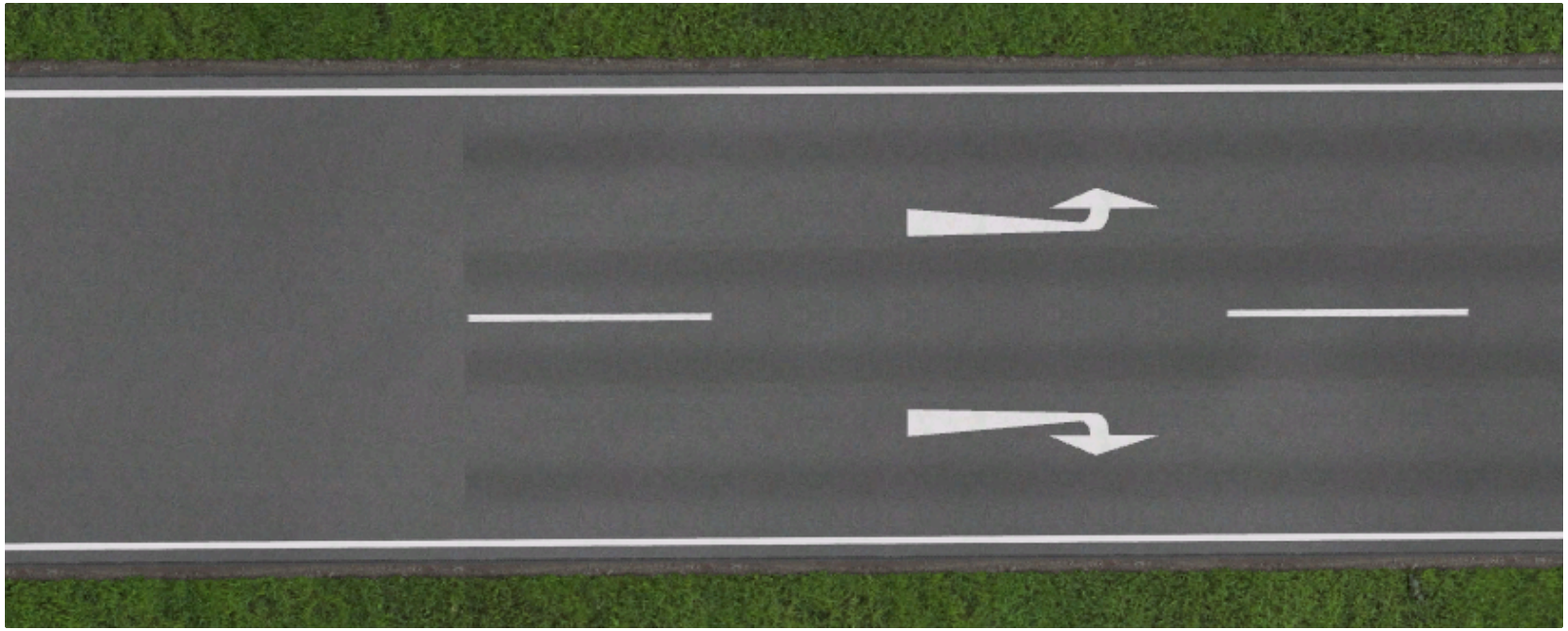

Figure 66. Example of a splitting driving lane

Figure 66 shows an example where a driving lane splits into two or more driving lanes abruptly. The width is non-zero.

Example where multiple predecessors and successors shall not be used:

- asam.net:xodr:1.4.0:road.lane.link.new_lane_appear: If a new lane appears besides, only the continuing lane shall be connected to the original lane, not the appearing lane.

Elements in UML model

`<link>` element

In ASAM OpenDRIVE, lane linkage is represented by the `<link>` element within the `<lane>` element.

```
UML class: t_road_lanes_laneSection_lcr_lane_link
XML tag:   <link> (Multiplicity: 0..1)
```

For links between lanes with an identical road reference line, the lane predecessor and successor information provide the IDs of lanes on the preceding or following lane section.

For links between lanes with different road reference line, the lane predecessor and successor information provide the IDs of lanes on the first or last lane section of the other road reference line depending on the contact point of the road linkage.

This element may only be omitted, if lanes end at a junction or have no physical link.

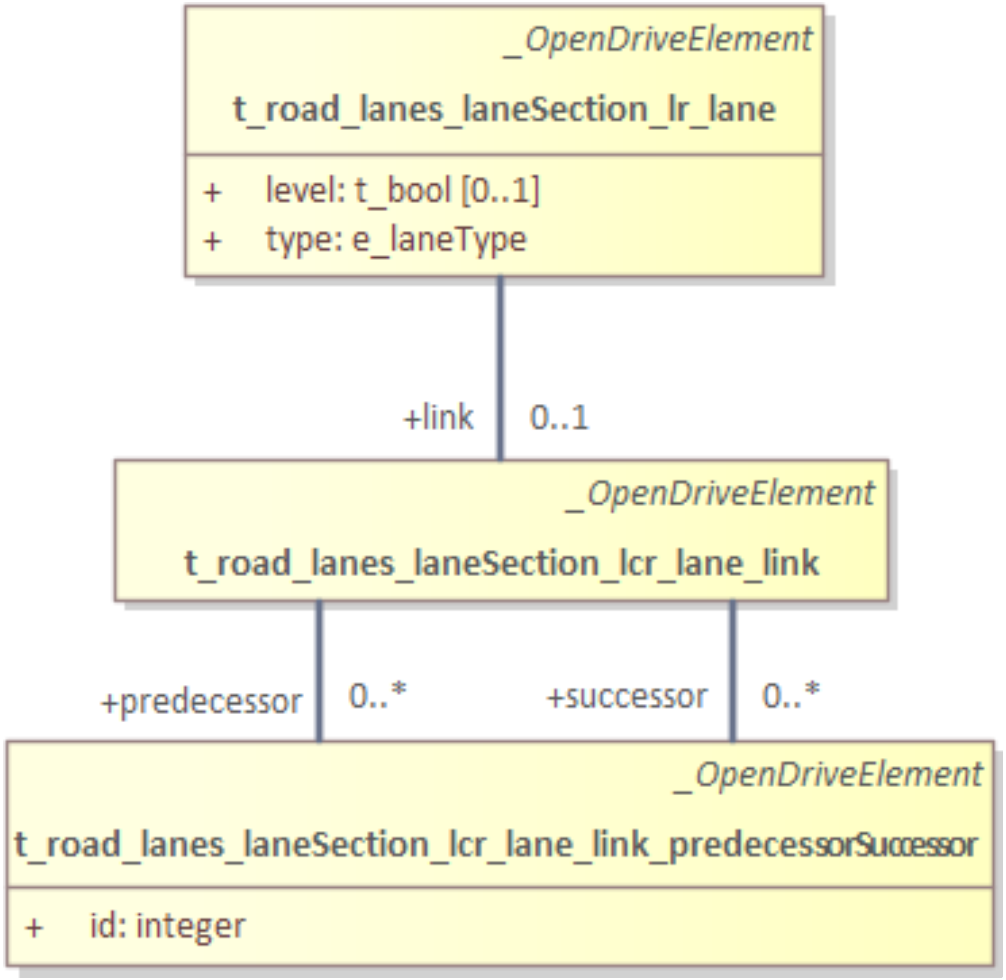

Figure 67. UML class diagram of the t_road_lanes_laneSection_lcr_lane_link element in the Lanes class

Figure 67 shows the UML class diagram of the t_road_lanes_laneSection_lcr_lane_link element in the ASAM OpenDRIVE Lanes class.

`<predecessor>` and `<successor>` elements

In ASAM OpenDRIVE, predecessors and successors are represented by the `<predecessor>` and `<successor>` elements within the `<link>` element.

```
UML class: t_road_lanes_laneSection_lcr_lane_link_predecessorSuccessor
XML tag:   <predecessor> (Multiplicity: 0..*)
XML tag:   <successor> (Multiplicity: 0..*)
```

Table 40. Attributes of the <predecessor> and <successor> elements

Name	Type	Use	Description
`id`	integer	required	ID of the preceding / succeeding linked lane

Rules

The following rules apply to lane linkage:

- A lane may have another lane as predecessor or successor.
- asam.net:xodr:1.4.0:road.lane.link.use_junctions: Two lanes shall only be linked if their linkage is clear. If the relationship to a predecessor or successor is ambiguous, junctions shall be used.
- asam.net:xodr:1.4.0:road.lane.link.multiple_connections: Multiple predecessors and successors shall be used if a lane is split abruptly or several lanes are merged abruptly. All lanes that are connected shall have a non-zero width at the connection point.
- asam.net:xodr:1.7.0:road.lane.link.zero_width_at_start: Lanes that have a width of zero at the beginning of the lane section shall have no `<predecessor>` element.
- asam.net:xodr:1.7.0:road.lane.link.zero_width_at_end: Lanes that have a width of zero at the end of the lane section shall have no `<successor>` element.

- asam.net:xodr:1.4.0:road.lane.link.no_link: The `<link>` element shall be omitted if the lane starts or ends in a junction or has no link.

Related topics

- Section 10.3, "Road linkage"
- Section 11.1, "Introduction to lanes"
- Section 12.1, "Introduction to junctions"